

Project Title

Exploratory data analysis with Coffee sales data and
MNIST data clustering.

Student Name : Suravi Mondal

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1. Abstract

In this project, Exploratory Data Analysis (EDA) was performed on both the Coffee Sales and MNIST datasets to better understand their underlying patterns. For the Coffee Sales data, the focus was on identifying sales trends, customer preferences, and the products that contributed most to revenue. By visualizing sales across different time periods and product categories, it became easier to spot peak sales periods and business opportunities. The data was also cleaned and organized to ensure accurate analysis. Synthetic data was generated and displayed to demonstrate data simulation techniques. For the MNIST dataset, the goal was to explore handwritten digit images and understand how they differ from one another. K-Means clustering was applied to group similar digit images into clusters based on their features. The clustering performance was evaluated using the F1 Score to measure how well the clusters aligned with the actual digit labels. Finally, a confusion matrix was created to visualize clustering accuracy and identify patterns of misclassification. Overall, the analysis provided valuable insights into both business data and image data, demonstrating the importance of EDA as a first step in any data science project.

2. Introduction

Data analysis plays a vital role in extracting meaningful information from raw data and supporting informed decision-making across various domains. This project focuses on Exploratory Data Analysis (EDA) using Coffee Sales data and clustering analysis using the MNIST handwritten digits dataset. The purpose of the project is to understand data patterns, uncover hidden trends, and apply machine learning techniques to solve real-world analytical problems.

The Coffee Sales dataset was selected to demonstrate how EDA can be used in a business context. Through data cleaning, statistical analysis, and visualization, insights such as sales trends, customer preferences, product performance, and seasonal variations can be identified. These insights help organizations optimize inventory management, improve marketing strategies, and enhance business performance.

The MNIST dataset, one of the most widely used benchmark datasets in machine learning, was chosen to explore unsupervised learning techniques. By applying K-Means clustering, handwritten digit images are grouped based on visual similarity without using their labels. The clustering results are then evaluated using performance metrics such as the F1 Score and Confusion Matrix to assess their effectiveness.

The project utilizes Python and several data science libraries, including Pandas for data manipulation, NumPy for numerical computations, `sklearn.datasets.load_digits` for loading the handwritten digits dataset used in clustering, `sklearn.cluster` for performing K-Means clustering, and `sklearn.metrics` for evaluating clustering performance using metrics such as the F1 Score and Confusion Matrix. These libraries provide a powerful framework for performing data analysis and machine learning tasks efficiently.

The procedure followed in this project includes importing libraries, loading datasets, conducting initial data inspection, generating descriptive statistics, converting date fields into

datetime format, creating additional temporal features such as month and year, performing sales aggregation and analysis, generating synthetic data, loading and preprocessing the MNIST dataset, applying K-Means clustering, and evaluating the clustering results.

Overall, this project demonstrates the importance of exploratory data analysis and unsupervised machine learning in extracting valuable insights from structured and image-based datasets. It provides practical experience in data preprocessing, visualization, statistical analysis, clustering, and performance evaluation, making it highly relevant to modern data science and analytics applications.

Key Learnings from the Summer Internship at IDEAS, ISI Kolkata

During the internship, I gained foundational knowledge and practical exposure in the field of Data Science. The key topics I learned are listed below:

Python Programming

- ✚ Data types and variables
- ✚ Lists and loops
- ✚ Python data structures
- ✚ Functions and classes
- ✚ Object-Oriented Programming (OOP)
- ✚ Writing and executing Python programs

Python Libraries

- ✚ Introduction to NumPy
- ✚ Introduction to Pandas
- ✚ Data manipulation and analysis using NumPy and Pandas
- ✚ Implementing programs using these libraries

Machine Learning Fundamentals

- ✚ Introduction to Machine Learning and its applications
- ✚ Machine Learning workflow
- ✚ Data collection and data preparation
- ✚ Model training and pattern recognition
- ✚ Supervised Learning
- ✚ Unsupervised Learning
- ✚ Reinforcement Learning
- ✚ Real-world applications of Machine Learning

Large Language Models (LLMs)

- ✚ Introduction to LLMs
- ✚ Basic concepts and working principles of Large Language Models
- ✚ Applications of LLMs in modern AI systems

Professional Development

- ✚ Communication skills enhancement
- ✚ Understanding the Data Science industry and its work environment

Overall, this internship provided me with a strong foundation in Python, Machine Learning, and LLM fundamentals, while also giving me valuable insight into the practical applications of Data Science in industry.

3. Project Objective

Objectives of the Project :

- To perform Exploratory Data Analysis (EDA) on the coffee sales dataset and identify sales trends, customer preferences, and product performance.
- To use data visualization and summary statistics to better understand the patterns and characteristics present in the coffee sales data.
- To apply K-Means clustering on the MNIST handwritten digit dataset to group similar digit images based on their features without using predefined labels.
- To demonstrate how unsupervised machine learning can uncover hidden patterns and similarities in image data.
- To evaluate the clustering performance using metrics such as the F1-Score and Confusion Matrix, and assess how well the clusters correspond to the actual digit classes.
- No sample survey was conducted for this project. The analysis is based entirely on secondary datasets Coffee Sales (provided beforehand) and MNIST.

4. Methodology

1. Data Loading and Initial Exploration

The analysis begins by importing the Pandas library and loading the dataset `Coffee_sales.csv` into a DataFrame named `coffee_data`. The first five rows of the dataset will be displayed using the `head()` function to verify that the data has been loaded correctly and to gain an initial understanding of its structure and contents.

Tools Used

✚ Pandas (pd.read_csv())

✚ head()

2. Data Inspection and Quality Assessment

A preliminary inspection of the dataset will be conducted to understand its dimensions and identify potential data quality issues. The number of rows and columns will be determined using the shape attribute. Duplicate records will be identified using the duplicated() function, while missing values across the entire dataset will be calculated using isnull().sum().sum().

Tools Used

✚ .shape

✚ duplicated()

✚ isnull()

3. Descriptive Statistical Analysis

Descriptive statistics will be generated for all numerical variables in the dataset. Measures such as count, mean, standard deviation, minimum value, maximum value, and quartiles will be computed to summarize the distribution of numerical features.

Tools Used

✚ describe()

4. Date-Time Transformation and Feature creation

The Date column, originally stored as an object data type, will be converted into a datetime format using Pandas datetime functionality. Additional temporal features will then be extracted, Month: numeric representation of the month and Year: numeric representation of the year.

After transformation, the data types of all columns will be reviewed to confirm successful conversion.

Tools Used

✚ pd.to_datetime()

✚ .dt.month

✚ .dt.year

✚ dtypes

5. Data Aggregation and Analysis

Several aggregation and grouping operations will be performed to derive business insights from the coffee sales data.

5.1 Average Sales per Year

The dataset will be grouped by the Year column, and the average value of the money column will be calculated for each year.

Method:

- ✚ Group by Year
- ✚ Apply mean() on money

5.2 Maximum Sales by Month Name

The maximum sales amount (money) will be identified for each month using the Month_name column.

Method:

- ✚ Group by Month_name
- ✚ Apply max() on money

5.3 Total Sales by Coffee Name

Total revenue generated by each coffee product will be computed.

Method:

- ✚ Group by coffee_name
- ✚ Apply sum() on money

5.4 Unique Coffee Types

The number and names of unique coffee products available in the dataset will be determined.

Method:

- ✚ Use unique() and len(unique_coffee_names)

5.5 Average Sales by Time of Day

The average sales generated during different periods of the day (Morning, Afternoon, Night) will be calculated.

Method:

- ✚ Group by Time_of_Day
- ✚ Apply mean() on money

6. Synthetic Data Generation

A synthetic dataset named synthetic_df will be created for demonstration purposes. The dataset will contain a date column consisting of 100 consecutive dates beginning on January 1, 2023 and a value column containing 100 randomly generated numerical values.

Tools Used

- ✚ pd.date_range()

✚ numpy.random

7. MNIST Dataset Loading and Preparation

The MNIST handwritten digit dataset will be loaded using Sklearn's `load_digits()` function. The dataset will be separated into *Features (X)*: Pixel intensity values and *Target Labels (y)*: Digit classes (0–9).

The dimensions of both arrays will be examined to confirm successful loading.

Tools Used

✚ `load_digits()`

✚ `.data`

✚ `.target`

✚ `.shape`

8. K-Means Clustering

An unsupervised machine learning approach will be applied using the K-Means clustering algorithm. Since the dataset contains ten digit classes (0–9), the number of clusters will be set to 10.

Procedure

1. Initialize the K-Means model.
2. Fit the model using feature matrix X.
3. Obtain cluster assignments for each sample.
4. Store cluster labels in `kmeans_labels`.

Tools Used

✚ `sklearn.cluster.KMeans`

9. Clustering Performance Evaluation Using F1 Score

Because K-Means cluster labels are assigned arbitrarily, direct comparison with true labels is not meaningful. Therefore, each cluster will be mapped to the most frequently occurring true digit label within that cluster.

Tools Used

✚ `scipy.stats.mode()`

✚ `sklearn.metrics.f1_score`

10. Confusion Matrix Analysis

A confusion matrix will be generated to evaluate the alignment between the mapped cluster labels and the true digit labels. The matrix provides a detailed class-by-class comparison, indicating correctly and incorrectly clustered samples.

Tools Used

✚ sklearn.metrics.confusion_matrix

➤ Libraries Used

The implementation will be carried out in Python using the following libraries:

✚ Pandas

✚ NumPy

✚ Sklearn

✚ Scipy

These libraries provide efficient tools for data manipulation, statistical analysis, machine learning, and performance evaluation.

➤ The required GitHub Link :

https://github.com/suravi-mondal/02-Exploratory_data_analysis_with_coffee_sales_data_Spring_2026.ipynb

5. Data Analysis and Results

1. Data Loading and Initial Inspection

The Pandas library was imported as pd, and the dataset Coffee_sales.csv was loaded into a DataFrame named coffee_data. The first five rows were displayed to verify successful loading and inspect the structure of the dataset.

Findings

- Dataset successfully loaded into a Pandas DataFrame.
- First five records showed information related to coffee sales transactions including date, coffee type, sales , and time-related attributes.

2. Dataset Inspection

A preliminary inspection was conducted to understand the overall quality and structure of the dataset.

Results

<i>metric</i>	<i>value</i>
Number of Rows	3547
Number of Columns	11
Duplicate Rows	0
Total Missing Values	0

Interpretation

- ✚ The dataset contains 3547 observations and 11 variables.
- ✚ A total of 0 duplicate records were identified.
- ✚ The dataset contains 0 missing values.

3. Descriptive Statistics

Descriptive statistics were generated for all numerical variables using the describe() function.

Results

<i>Statistic</i>	<i>hour_of_day</i>	<i>money</i>	<i>Weekdaysort</i>	<i>Monthsort</i>
<i>Count</i>	3547.000000	3547.000000	3547.000000	3547.000000
<i>Mean</i>	14.185791	31.645216	3.845785	6.453905
<i>Std Dev</i>	4.234010	4.877754	1.971501	3.500754
<i>Minimum</i>	6.000000	18.120000	1.000000	1.000000
<i>25th Percentile</i>	10.000000	27.920000	2.000000	3.000000
<i>Median (50%)</i>	14.000000	32.820000	4.000000	7.000000
<i>75th Percentile</i>	18.000000	35.760000	6.000000	10.000000
<i>Maximum</i>	22.000000	38.700000	7.000000	12.000000

Interpretation

- ✚ The average sales amount was 31.645216
- ✚ The highest recorded sale was 38.700000
- ✚ The minimum sale amount was 18.120000

4. Date Conversion and Feature Engineering

The Date column was converted from an object datatype to a datetime datatype using pd.to_datetime(). Two new columns were created, Month : month number extracted from the date. Year : year extracted as an integer.

Updated Data Types

<i>Column</i>	<i>Data Type</i>
<i>Date</i>	datetime64[ns]
<i>Month</i>	int64
<i>Year</i>	int64
<i>hour_of_day</i>	int64
<i>cash_type</i>	object
<i>money</i>	float64
<i>coffee_name</i>	object
<i>Time_of_Day</i>	object
<i>Weekday</i>	object
<i>Month_name</i>	object
<i>Weekdaysort</i>	int64
<i>Monthsort</i>	int64
<i>Time</i>	object

Interpretation

- ✚ Converting the date column enabled time series analysis such as monthly and yearly sales trends.

5. Exploratory Data Analysis

5.1 Average Sales per Year

The dataset was grouped by the Year column and the average sales (money) was calculated.

Results

<i>Year</i>	<i>Average Sales</i>
2024	31.737634
2025	31.390011

Interpretation

- ✚ More or less the years had very close average sales, indicating the business performance was pretty steady over the given period of time.

5.2 Maximum Sales by Month Name

The maximum sales value for each month was calculated.

Results

<i>Month Name</i>	<i>Maximum Sale</i>
January	35.76
February	35.76
March	38.70
April	38.70
May	37.72
June	37.72
July	37.72
September	35.76
October	35.76
November	35.76
December	35.76

Interpretation

- ✚ The month with the highest sale value was March and April.

5.3 Total Sales by Coffee Name

Total revenue generated by each coffee type was calculated.

Results

<i>Coffee Name</i>	<i>Total Sales</i>
Americano	14650.26
Americano with Milk	24751.12

<i>Cappuccino</i>	17439.14
<i>Cocoa</i>	8521.16
<i>Cortado</i>	7384.86
<i>Espresso</i>	2690.28
<i>Hot Chocolate</i>	9933.46
<i>Latte</i>	26875.30

Interpretation

✚ The best-performing coffee product was Latte, generating total sales of 26875.30.

5.4 Unique Coffee Types

The number of unique coffee products was determined.

Results

- Number of unique coffee types : 8
- Coffee names : ['Latte', 'Hot Chocolate', 'Americano', 'Americano with Milk', 'Cocoa', 'Cortado', 'Espresso', 'Cappuccino']

Interpretation

✚ The dataset contains a diverse range of coffee products, allowing comparison of product-level performance.

5.5 Average Sales by Time of Day

Average sales were calculated for each period of the day.

Results

<i>Time of Day</i>	<i>Average Sales</i>
Morning	30.422693
Afternoon	31.643187
Night	32.890904

Interpretation

✚ The highest average sales occurred during night, indicating peak customer purchasing behaviour.

6. Synthetic DataFrame Creation

A synthetic dataset named `synthetic_df` was generated.

Structure

- ✚ 100 consecutive dates beginning from 2023-01-01.
- ✚ 100 randomly generated numerical values.

Sample Structure

<i>Si. No.</i>	<i>date</i>	<i>value</i>
----------------	-------------	--------------

0	2023-01-01	0.745450
1	2023-01-02	0.055390
2	2023-01-03	0.407229
3	2023-01-04	0.364728
4	2023-01-05	0.249515
...	...	...
95	2023-04-06	0.661361
96	2023-04-07	0.768852
97	2023-04-08	0.928411
98	2023-04-09	0.481001
99	2023-04-10	0.904225

Interpretation

- ✚ The synthetic dataset can be used for time-series experimentation, trend analysis, and machine learning demonstrations.

7. MNIST Dataset Loading

The MNIST Digits dataset was loaded from `sklearn.datasets.load_digits()`.

Results

- The shape of X is : (1797, 64)
- The shape of y is : (1797,)

Interpretation

Each sample contains pixel-based features representing handwritten digits from 0–9.

8. K-Means Clustering

K-Means clustering was applied to the feature matrix using:

- Number of clusters = 10
- Random initialization
- Cluster assignments were stored in the variable `kmeans_labels`.

Interpretation

- ✚ The model attempted to partition the handwritten digit images into ten clusters corresponding to the ten digit classes.

9. F1 Score Evaluation

Cluster labels were mapped to the most frequent true digit label within each cluster since K-Means produces arbitrary cluster labels.

The macro-averaged F1 score was then computed.

Result

- The calculated F1 score is : 0.8627854786352394

Interpretation

- ✚ F1 scores closer to 1 indicate stronger alignment between clusters and actual digit classes.
- ✚ A score of 0.8627854786352394 suggests good clustering performance.

10. Confusion Matrix

A confusion matrix was generated comparing the mapped cluster labels against the true digit labels.

Confusion Matrix

```
[[ 0  0  0  0  2 176  0  0  0  0]
 [ 0  1  2 23  0  0  1 130  3 22]
 [ 2  0  0 146  0  1  12  6  9  1]
 [ 7  2  0  1  0  0 159  0  6  8]
 [ 5  0  0  0 159  0  0  13  4  0]
 [ 0 137  2  0  2  0  0  0  0 41]
 [ 0  0 176  0  0  1  0  3  1  0]
 [169  0  0  0  0  0  0  0  10  0]
 [ 2  2  1  3  0  0  0  14 136 16]
 [ 8  4  0  0  3  0  1  1  3 160]]
```

Interpretation

- ✚ Diagonal values represent correctly grouped digits.
- ✚ Higher concentration along the diagonal signifies better clustering quality.

6. Conclusion

The coffee sales analysis provided valuable insights into sales performance and customer purchasing patterns. Descriptive statistical analysis showed that the average sales amount was 31.65, with a maximum sale of 38.70 and a minimum sale of 18.12. Analysis of yearly sales revealed that the average sales across different years were very similar, indicating that the business maintained relatively stable performance over the observed period.

Time-based analysis showed that March and April recorded the highest sales values, suggesting stronger customer demand during these months. Product-wise analysis identified Latte as the best-performing coffee product, generating total sales of 26,875.30, making it the most popular item in the dataset. Furthermore, sales by time of day indicated that the night period had the highest average sales, highlighting it as the peak customer purchasing period.

For the machine learning component, the MNIST handwritten digit dataset was clustered using the K-Means algorithm with 10 clusters. After mapping cluster labels to their corresponding digit classes, the clustering performance was evaluated using the macro-averaged F1 score. The model achieved an F1 score of 0.863, indicating that K-Means was able to group handwritten digit images with a relatively high level of accuracy. The confusion matrix further demonstrated that most digits were clustered correctly, although some visually similar digits were occasionally grouped together.

Overall, this project successfully demonstrated the practical application of data preprocessing, exploratory data analysis, feature engineering, clustering, and model evaluation using Python's Pandas and Scikit-Learn libraries. As a future enhancement, additional business-related variables such as customer demographics, weather conditions, promotional campaigns, or holiday seasons could be incorporated into the coffee sales dataset to gain deeper insights into the factors influencing sales performance. Similarly, further experimentation with different numbers of clusters and feature selection techniques could help improve the clustering quality of the MNIST dataset.

7. APPENDICES

Appendix A: References

The following resources were consulted during the completion of this project:

- ✚ I took reference from the online theory classes that were taken in the first two weeks of our internship program.
- ✚ W3Schools Python Tutorial :
<https://www.w3schools.com/python/>
- ✚ GeeksforGeeks – Python Programming and Machine Learning Resources :
<https://www.geeksforgeeks.org/>
- ✚ Kaggle – Data Science and Machine Learning Resources :
<https://www.kaggle.com/>

Dataset References :

- ✚ Coffee Sales Dataset (provided beforehand).
- ✚ MNIST Digits Dataset accessed through `sklearn.datasets.load_digits`.

Appendix B: Survey Questionnaire

No survey or questionnaire was conducted as part of this project. The project was based entirely on secondary datasets and machine learning analysis.

Appendix C: GitHub Repository

The source code developed for this project is available at GitHub Repository:
https://github.com/suravi-mondal/02-Exploratory_data_analysis_with_coffee_sales_data_Spring_2026.ipynb

Appendix D: Supporting Documents

The following project documents have been uploaded and are available through GitHub Repository :